



PID CONTROL

STRUCTURE, TUNING AND IMPLEMENTATION

STEFAN BROCK

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Poznan University of Technology

Institute of Robotics and Machine Intelligence

The Erasmus Programme (European Community Action Scheme for the Mobility of University Students) is a European Union student and teacher exchange programme.

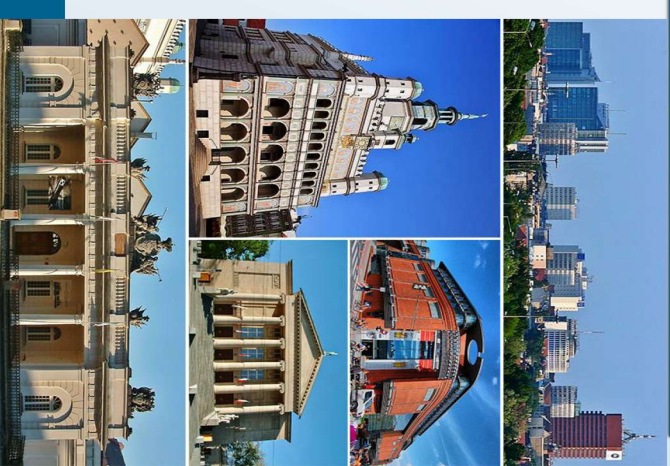


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Poznań

1. Poznan is a city on the Warta river in west-central Poland, with a population of 550 thousand - the largest metropolitan area in western Poland.
2. Poznan is among the oldest cities in Poland
3. The city hosts regular international trade fairs and international conferences.

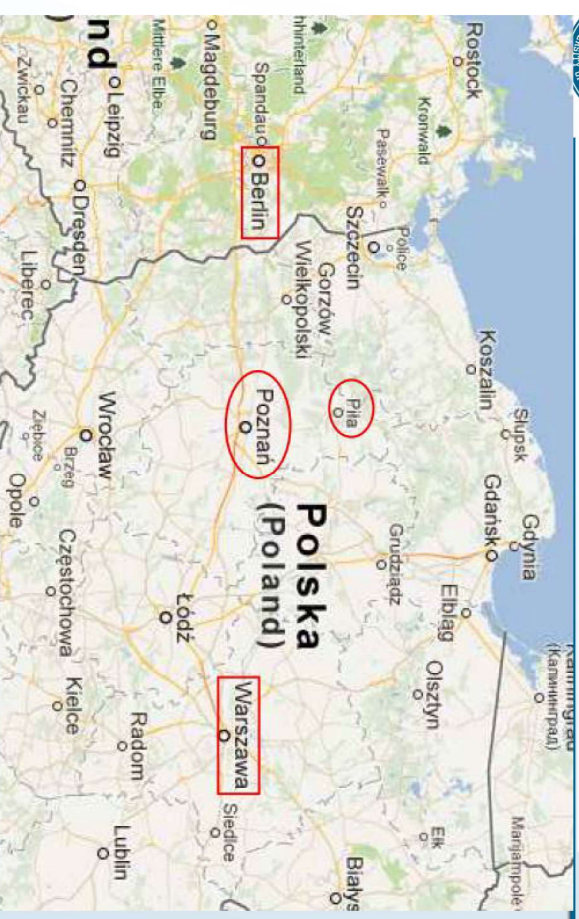


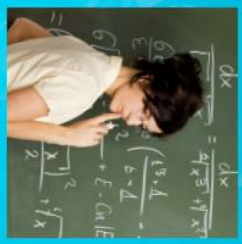
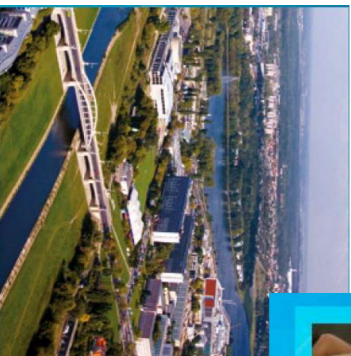
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POZNAN UNIVERSITY OF TECHNOLOGY IN NUMBERS

- ten faculties
- almost 20 000 students
- 26 fields of study
- over 50 post-graduate courses
- more than 1,000 academic staff



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Stefan Brock

- Scientific:
 - Robust control of electrical drives and motion systems
- Didactic:
 - Control of electrical drives
 - Machine learning, fuzzy logic control and neural networks control
 - Industrial control systems: PLC, DCS and SCADA

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Incorrect forecasts for PID controllers

- ▲ 1982: The ASEA Novatune Team 1982 (Novatune is a useful general digital control law with adaptation):

PID Control will soon be obsolete

*) B. Bernhardsson, K.J. Åström Control System Design – PID Control, 2007

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- ▲ 1989: Conference on Model Predictive Control:

Using a PI controller is like driving a car only looking at the rear view mirror: It will soon be replaced by Model Predictive Control.

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Using a PI controller is like driving a car only looking at the rear view mirror: It will soon be replaced by Model Predictive Control.
- ▲ 2002: Desborough and Miller (Honeywell):
Based on a survey of over 11000 controllers in the refining, chemicals and pulp and paper industries, 98% of regulatory controllers utilise PID feedback

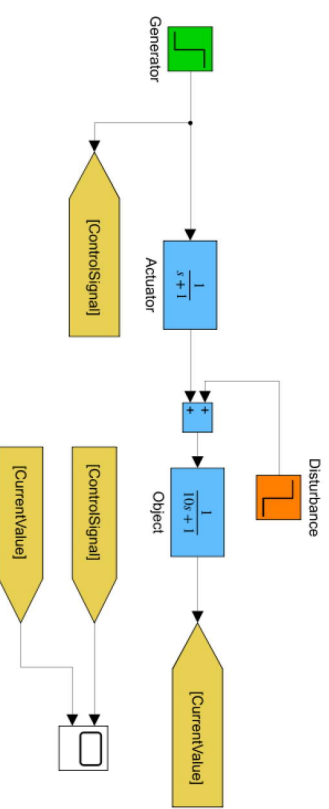
*) B. Bernhardsson, K.J. Åström Control System Design – PID Control, 2007

Why it is worthwhile to consider PI/PID controllers

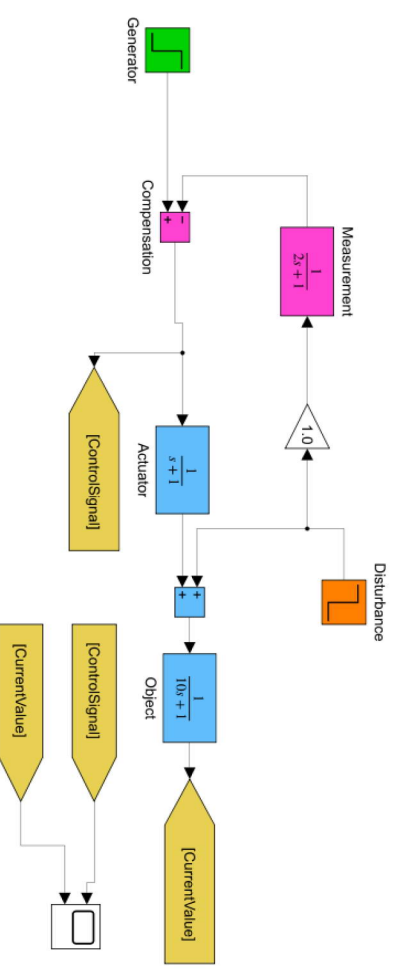
- Most regulators in practical control systems (not only drive systems) are PI/PID controllers.
- Unlike other controllers, they can easily be set up experimentally, even if the control unit is not fully understood; the rules of procedure are (should be) well known by the engineers.
- Relatively easy: Self-adjustment and adaptation
- The results obtained in the system with PI/PID controller can be used as a basis for comparison with other control methods.

Typical structure in control systems

– open loop control

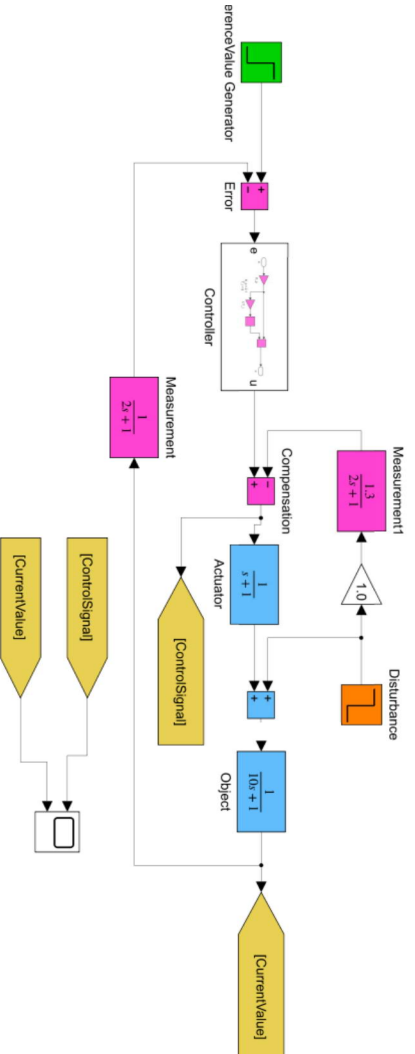
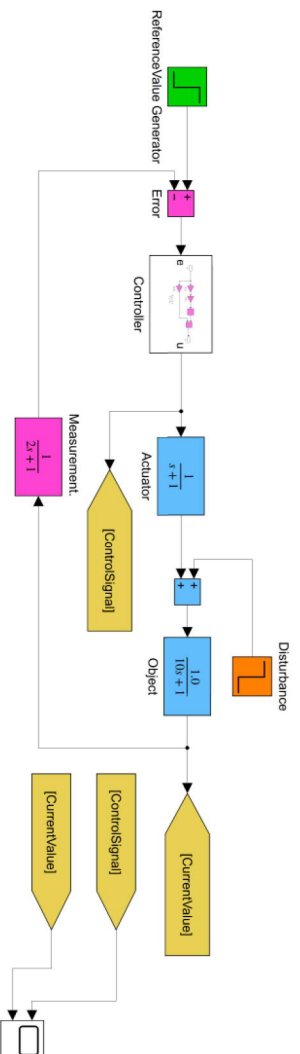


Typical structure in control systems – open loop control with compensation



Typical structure in control systems

– close loop control

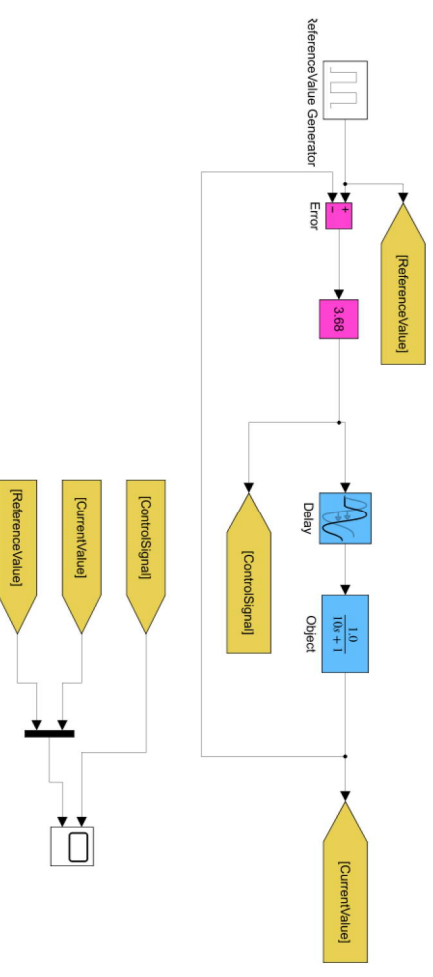


Typical structure in control systems

– close loop control with compensation

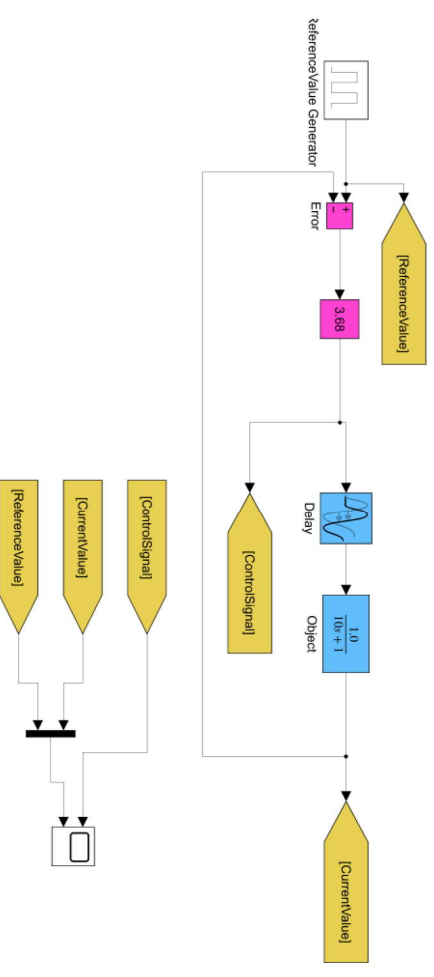
Rating the quality of the control system - simple

object with P controller



Rating the quality of the control system - simple

object with P controller



Rating the quality of the control system - simple object with P controller

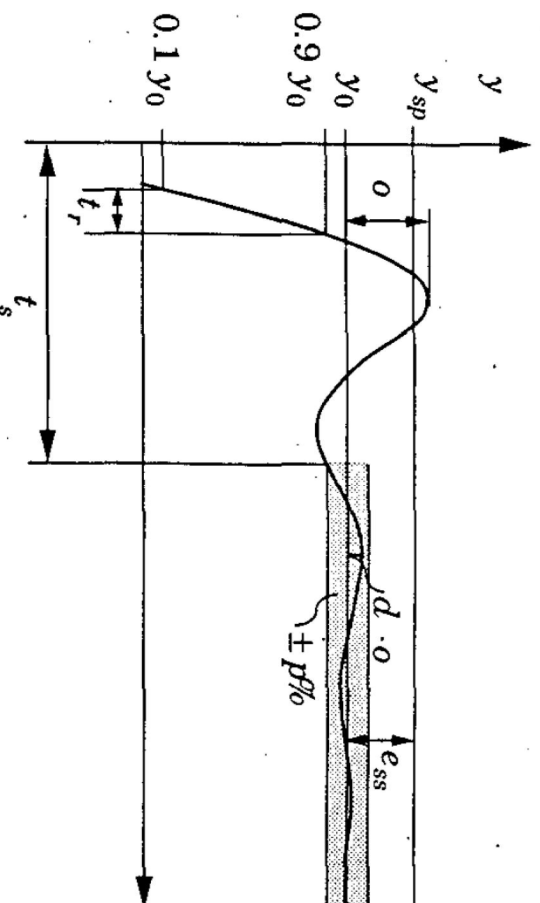
```
%% define object and controller
s=tf('s');
Object=exp(-1*s)*(1/(10*s+1));
Controller=8.6;

%% calculate close loop system
OpenLoop=Controller*Object;
CloseLoop=feedback(OpenLoop,1);

%% step response of close loop system
figure()
step(CloseLoop);

%% Bode characteristic of open loop system
figure()
bode(OpenLoop);
```

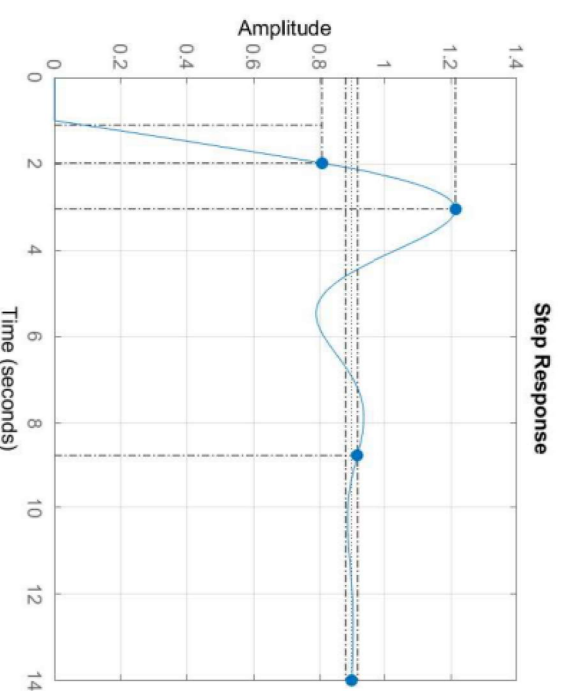
Parameters of the step response



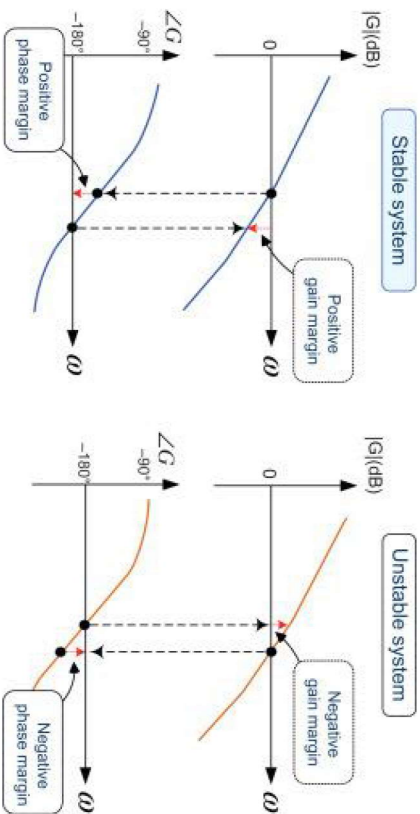
Parameters of the step response

1. Rise Time: the time it takes for the plant output y to rise beyond 90% of the desired level for the first time.
2. Overshoot: how much the the peak level is higher than the steady state, normalized against the steady state.
3. Settling Time: the time it takes for the system to converge to its steady state.
4. Steady-state Error: the difference between the steady-state output and the desired output.

Parameters of the step response



Bode characteristic of Open Loop

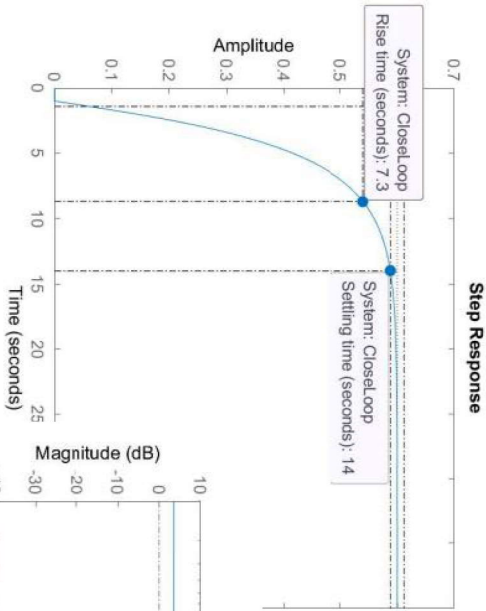
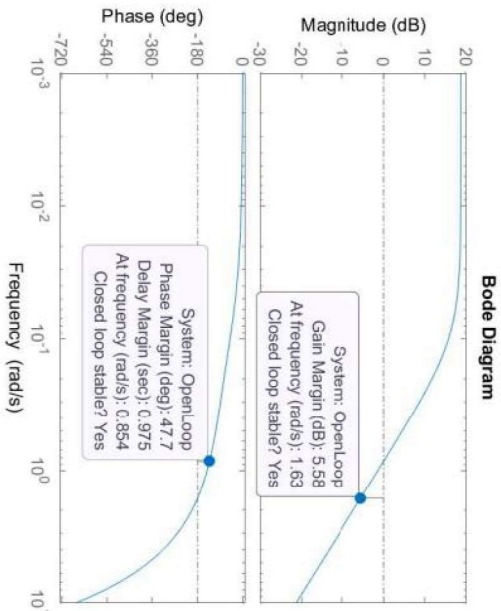


Gain margin is the amount of gain variance required to make the loop gain unity at the frequency, where the phase angle is -180°

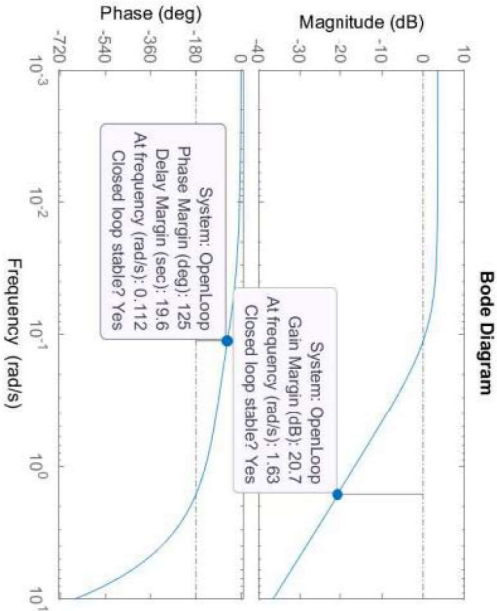
Phase margin is the difference between the phase of the response and -180° when the loop gain is 1.0.

Usually, gain margins of 3 or more combined with phase margins between 30° and 60° result in reasonable tradeoffs between bandwidth and stability.

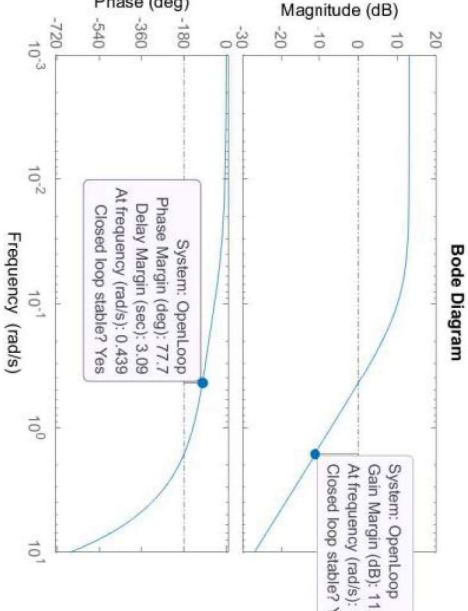
Bode characteristic of Open Loop



$$K_p = 1.5$$



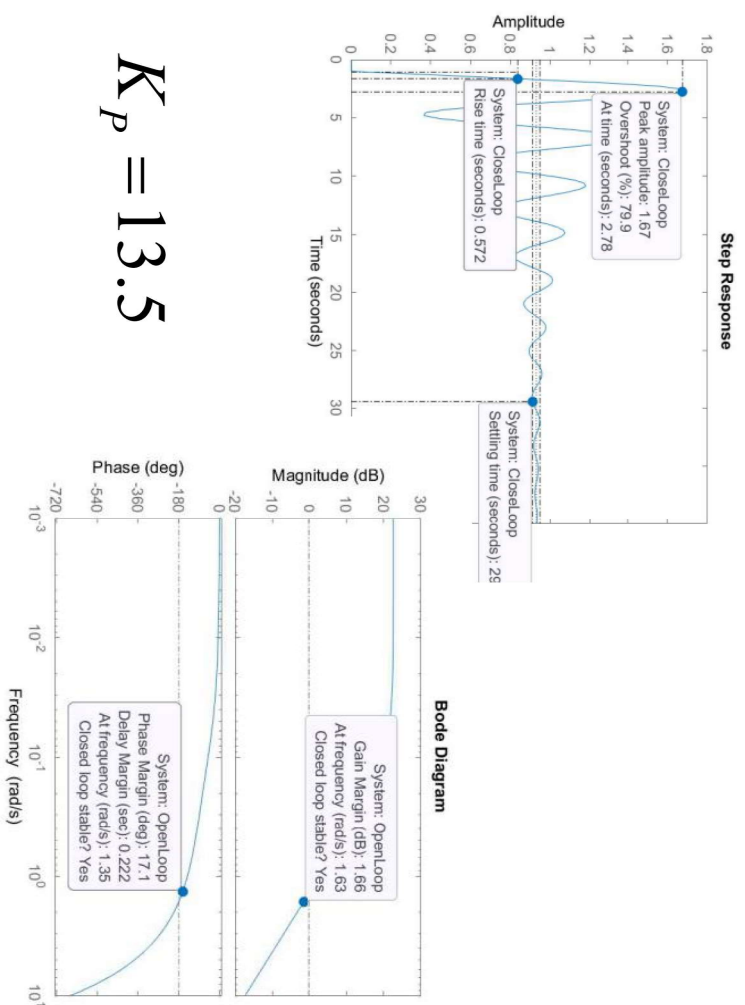
$$K_p = 4.5$$



Optimum Settings for Automatic Controllers

By J.G. ZIEGLER¹ and N. B. NICHOLS² • ROCHESTER, N. Y.

Published in 1942

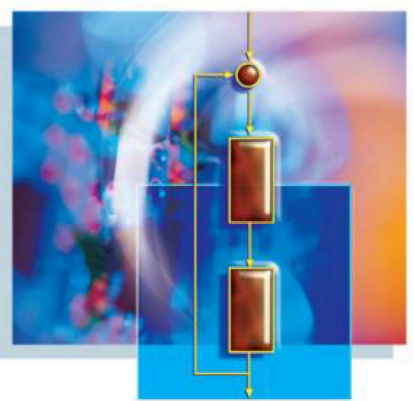


$$K_p = 13.5$$

HANDBOOK OF PI AND PID CONTROLLER TUNING RULES

3rd Edition

Aidan O'Dwyer



Imperial College Press

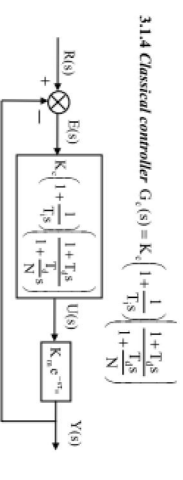
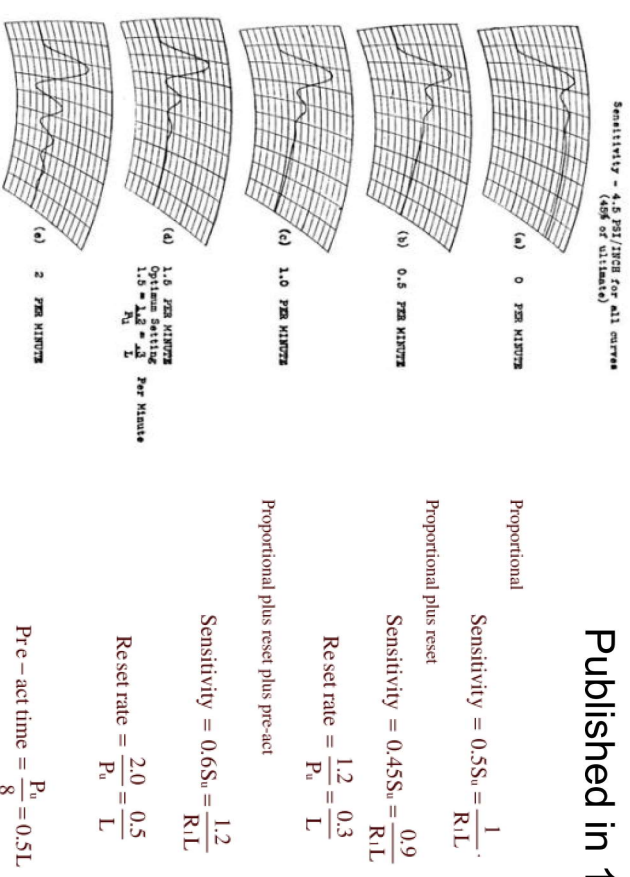


Table 5: PID controller tuning rules – delay model $G_p(s) = K_p e^{-\tau_m}$

Rule	K_c	T_i	T_d	Comment
Process reaction				
Hartree <i>et al.</i> (1937)	$\frac{1}{K_p \tau_m}$	$2.66 \tau_m$	τ_m	Representative results.
Model: Method 1	$\frac{1}{K_p \tau_m}$	$2.97 \tau_m$	$0.5 \tau_m$	

More than 600 pages of equation and tables, for various objects, controller structures and requirements.



The Ziegler – Nichols Tuning Rule

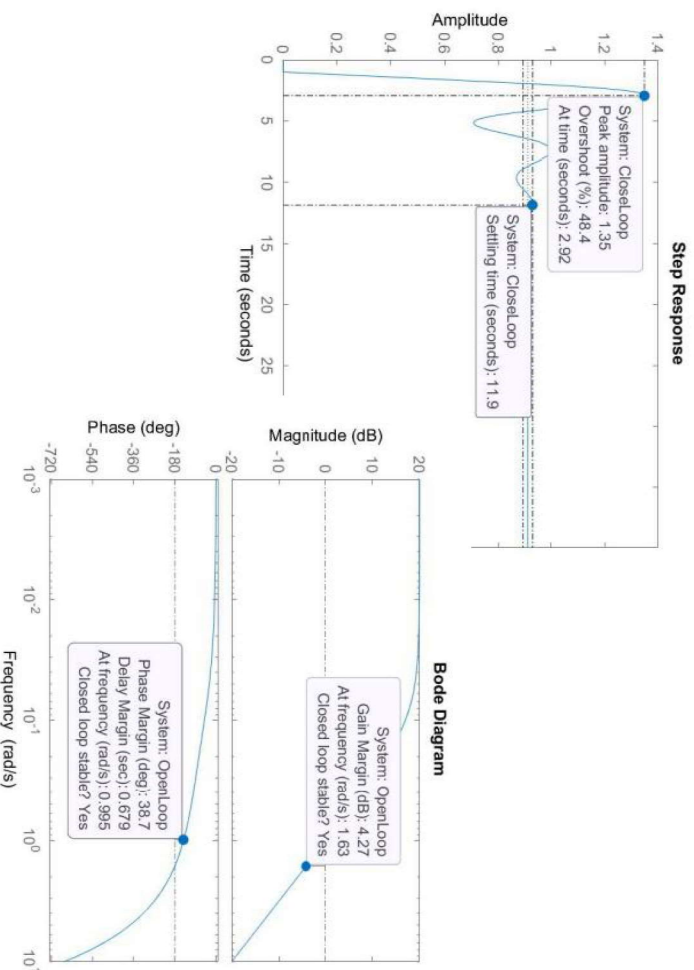
- For standard object KLT:

$$G_{ob}(s) = \frac{K e^{-sL}}{Ts + 1}$$

- P-type controller
- $$G_C(s) = K_p$$

- where
- $$K_p = \frac{1}{K} \cdot \frac{T}{L}$$

The Ziegler – Nichols Tuning Rule



Simple analytical tuning method

- We try to select the gain of the regulator P such that the closed loop system has the expected damping factor.

- Simplified object:

$$G_{ob}(s) = \frac{1}{1+sL} \cdot \frac{K}{1+sT}$$

- Controller:

$$G_c(s) = K_p$$

- Close loop system:

$$G_{cl}(s) = \frac{G_{ol}}{1+G_{ol}}$$

Simple analytical tuning method

- Close loop system:

$$G_{cl}(s) = \frac{G_{ol}}{1+G_{ol}}$$

$$= \frac{K_p \cdot K}{s^2 \cdot L \cdot T + s \cdot (L + T) + (1 + K_p \cdot K)}$$

- We will try to match the standard transmittance of a 2nd order system:

$$G_{sos}(s) = \frac{k \cdot \omega^2}{s^2 + 2\xi\omega s + \omega^2}$$

Simple analytical tuning method

- By varying the gain of the controller, we can obtain either the desired attenuation or the desired pulsation

$$2 \cdot \xi \cdot \omega = \frac{L+T}{L \cdot T}$$

$$\omega^2 = \frac{1+K_p \cdot K}{L \cdot T}$$

$$2 \cdot \xi \cdot \omega = \frac{L+T}{L \cdot T}$$

$$\omega^2 = \frac{1+K_p \cdot K}{L \cdot T}$$

Simple analytical tuning method

$$K_p = \frac{1}{K} \left(\frac{1}{4\xi^2} \frac{(L+T)^2}{LT} - 1 \right)$$

- For our example: L=1, T=10, K=1

$$\text{for } \xi = \frac{\sqrt{2}}{2} \Rightarrow K_p = 5.050$$

$$\text{for } \xi=1 \Rightarrow K_p = 2.025$$

- P-type controller is fast but does not eliminate steady-state error. An additional integral term is required.

PI controller

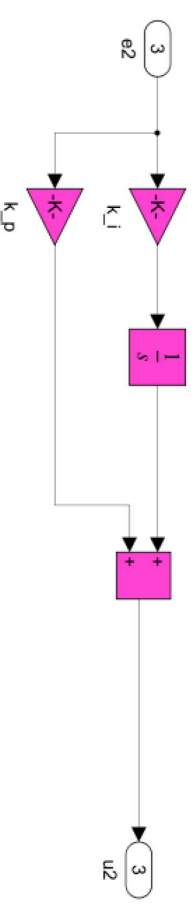
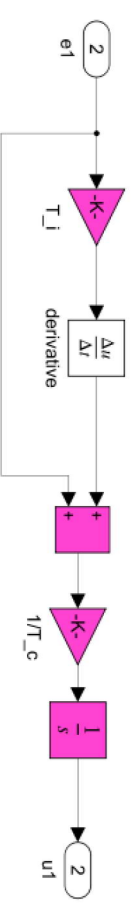
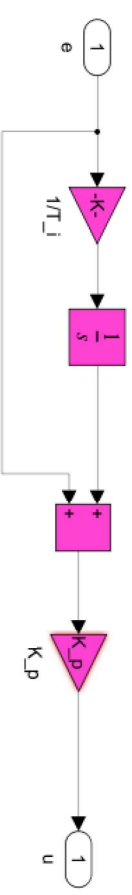
- The structure of the PI controller can be written in different forms: ideal, series, parallel. Other notations are also possible and have been found

$$G_c(s) = K_p \left(1 + \frac{1}{sT_i} \right) =$$

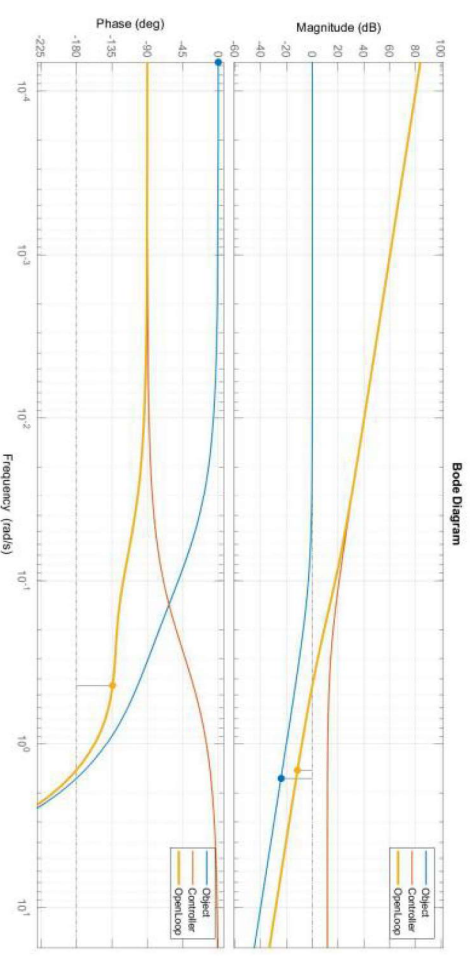
$$= (1 + sT_i) \cdot \frac{1}{sT_c} =$$

$$= k_p + k_i \frac{1}{s}$$

PI controller



PI controller



The Ziegler – Nichols Tuning Rule

- For standard object KLT:

$$G_{ob}(s) = \frac{Ke^{-sL}}{Ts + 1}$$

- PI-type controller

$$G_c(s) = K_p \left(1 + \frac{1}{sT_i} \right)$$

- where

$$K_p = \frac{0.9}{K} \cdot \frac{T}{L} \quad T_i = 3L$$

PID controller

A PI controller eliminates the steady-state error but requires a reduction in gain. An additional derivative term is required.

$$\begin{aligned} G_c(s) &= K_p \left(1 + \frac{1}{sT_i} + sT_d \right) = \\ &= K' \cdot \left(1 + \frac{1}{sT_i'} \right) \cdot (1 + sT_d') = \\ &= k_p + k_i \frac{1}{s} + k_d s \end{aligned}$$

The Ziegler – Nichols Tuning Rule

- For standard object KLT:

$$G_{ob}(s) = \frac{Ke^{-sL}}{Ts + 1}$$

- PID-type controller

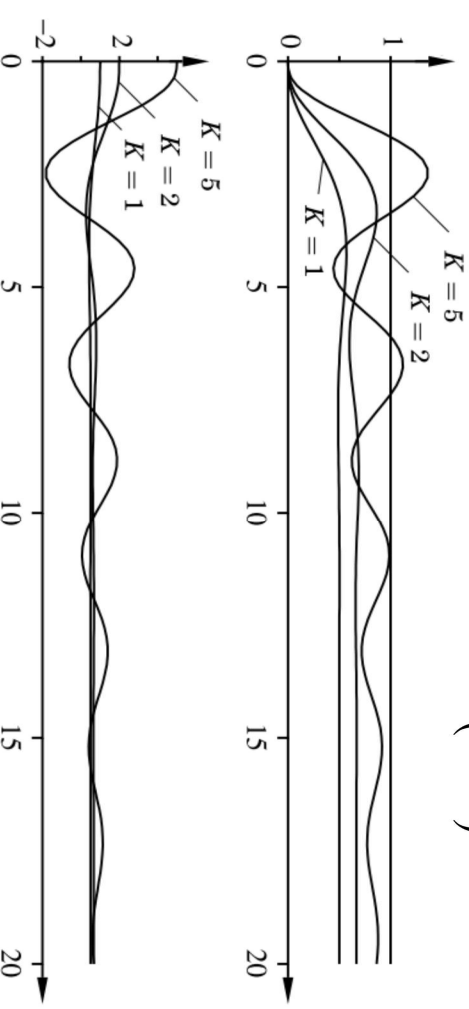
$$G_c(s) = K_p \left(1 + \frac{1}{sT_i} + sT_d \right)$$

- where

$$K_p = \frac{1.2}{K} \cdot \frac{T}{L} \quad T_i = 2L \quad T_d = 0.5L$$

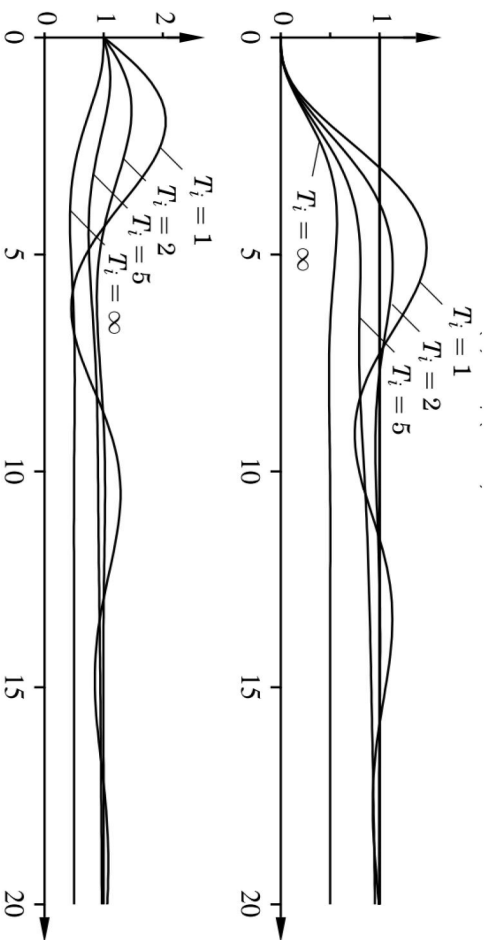
P Controller

$$G_{ob}(s) = \frac{1}{(s+1)^3}$$



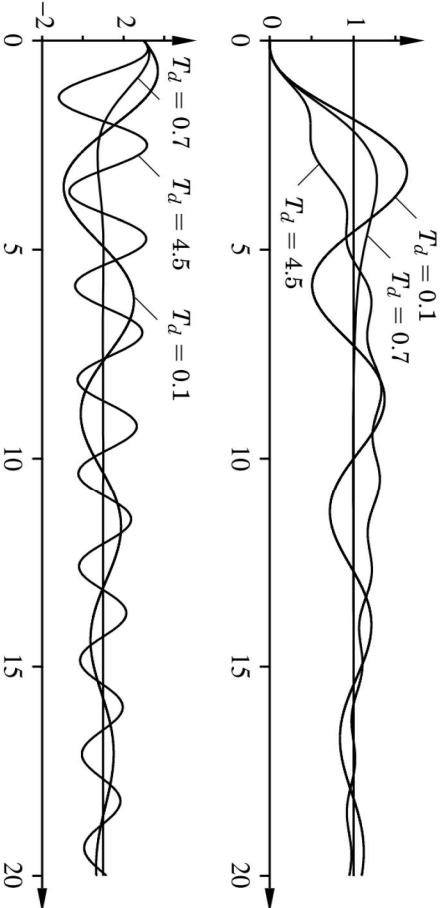
P controller reduces rise time and steady-state error, but increases oscillatory character

PI Controller K=1



I term eliminates steady-state error, but increases the oscillatory character

PID Controller. K=3, Ti=2



D term reduces oscillatory properties, but for large values of Td it excites oscillations.

Response	Rise Time	Overshoot	Settling Time	S-S Error
K_p	Decrease	Increase	NT	Decrease
K_i	Decrease	Increase	Increase	Eliminate
K_d	NT	Decrease	Decrease	NT

$$G_C(s) = k_p + k_i \frac{1}{s} + k_d s$$

When a PI controller is sufficient

- when the dynamic properties of the object are well approximated by a 1st order block (single tank level, mixing process,...);
- for higher-order objects, when the dynamic requirements are not sharp - the I part ensures the elimination of the steady-state error, the P part ensures adequate dynamics of the transition processes.

When a PID controller is sufficient

- when the dynamics of the object is well approximated by a 2nd order member;
- when the time constants vary considerably (for example temperature control); the D term improves the response of the control system;
- for higher-order objects, the limitation of the allowed gain of the P term is limited; the D member improves the damping and allows the gain to be increased.

Ziegler-Nichols Rules

- For standard object KLT: $G_{ob}(s) = \frac{Ke^{-sL}}{Ts + 1}$
- PID-type controller

controller	K_p	T_i	T_d
P	$\frac{1}{K} \frac{T}{L}$		
PI	$\frac{0.9}{K} \frac{T}{L}$	$3L$	
PID	$\frac{1.2}{K} \frac{T}{L}$	$2L$	$0.5L$

The Chien, Hrones and Reswick Method (CHR) – published in 1952

- Provides better damping than ZN
- Signal waveforms without overshoot or with 20% overshoot
- Other criteria for step response optimisation, other criteria for disturbance rejection

$$G_{ob}(s) = \frac{Ke^{-sL}}{Ts + 1} \quad G_c(s) = K_p \left(1 + \frac{1}{sT_i} + sT_d \right)$$

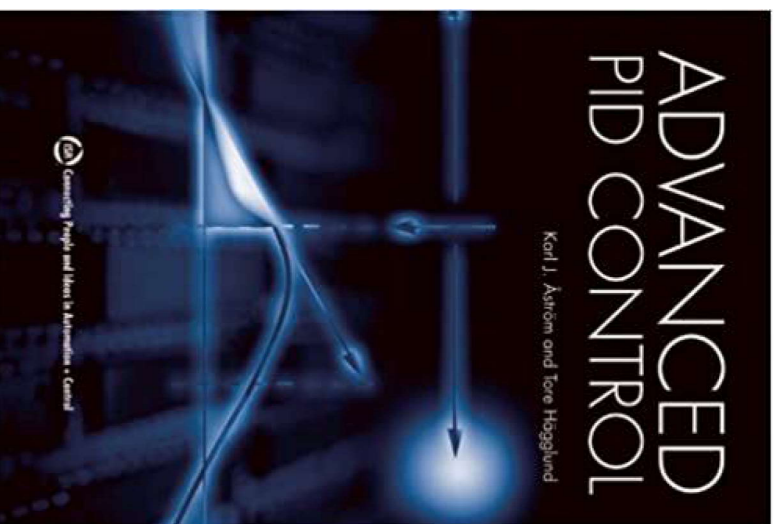
CHR Rules - setpoint step change

controller	K_p	T_i	T_d
P	$\frac{0.3}{K} \frac{T}{L}$		
PI	$\frac{0.35}{K} \frac{T}{L}$	$1.2T$	
PID	$\frac{0.6}{K} \frac{T}{L}$	T	$0.5L$

0% overshoot

20% overshoot

controller	K_p	T_i	T_d
P	$\frac{0.7}{K} \frac{T}{L}$		
PI	$\frac{0.6}{K} \frac{T}{L}$	T	
PID	$\frac{0.95}{K} \frac{T}{L}$	$1.4T$	$0.47L$



Advanced PID Control

Karl J. Åström Tore Hägglund

*Department of Automatic Control
Lund Institute of Technology
Lund University*

Published in 2005

The Astrom simplified tuning rule

- For integral object : $G_{ob}(s) = \frac{K_Y}{s} e^{-sL}$

- PID-type controller

$$G_c(s) = K_p \left(1 + \frac{1}{sT_i} + sT_d \right)$$

- where

$$K_p = \frac{0.45}{K_Y}$$

$$T_i = 8L$$

$$T_d = 0.5L$$

The Astrom simplified tuning rule

- For standard object KLT: $G_{ob}(s) = \frac{Ke^{-sL}}{Ts + 1}$

- PID-type controller
- $$G_c(s) = K_p \left(1 + \frac{1}{sT_i} + sT_d \right)$$

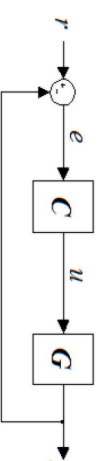
- where

$$K_p = \frac{1}{K} \left(0.2 + 0.45 \frac{T}{L} \right)$$

$$T_i = \frac{0.4L + 0.8T}{L + 0.1T}$$

$$T_d = \frac{0.5LT}{0.3L + T}$$

How **not** to think about controllers

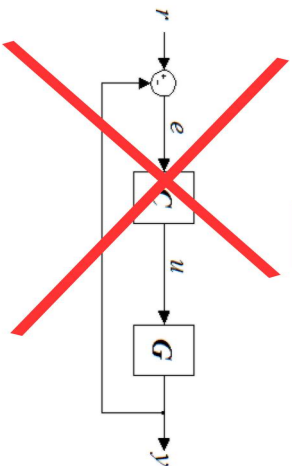


$$G_{ob}(s) = \frac{Y(s)}{U(s)}$$

$$C = f(G_{ob}(s))$$

*) K.J. Åström, H.I. Panagopoulos, T. Hägglund, Automatica vol. 34, no 5, 1998

How **not** to think about controllers



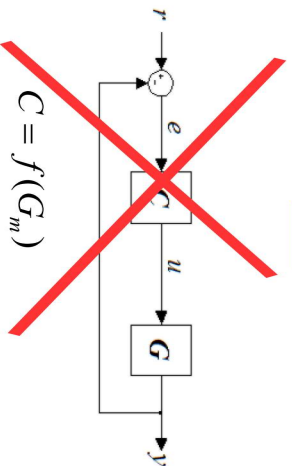
- incomplete structure
- only reaction $r \rightarrow y$
- no analysis of load and disturbance effects

$$G_{ob}(s) = \frac{Y(s)}{U(s)}$$

$$C = f(G_{ob}(s))$$

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How **not** to think about controllers



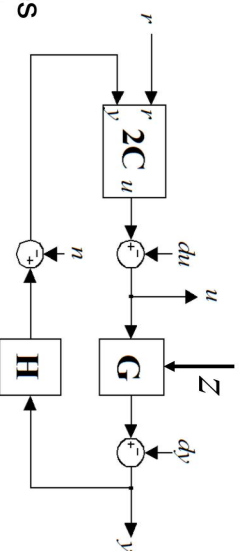
$$C = f(G_m)$$

- incomplete structure
- only reaction $r \rightarrow y$
- no analysis of load and disturbance effects

- two degrees of freedom (2DOF)

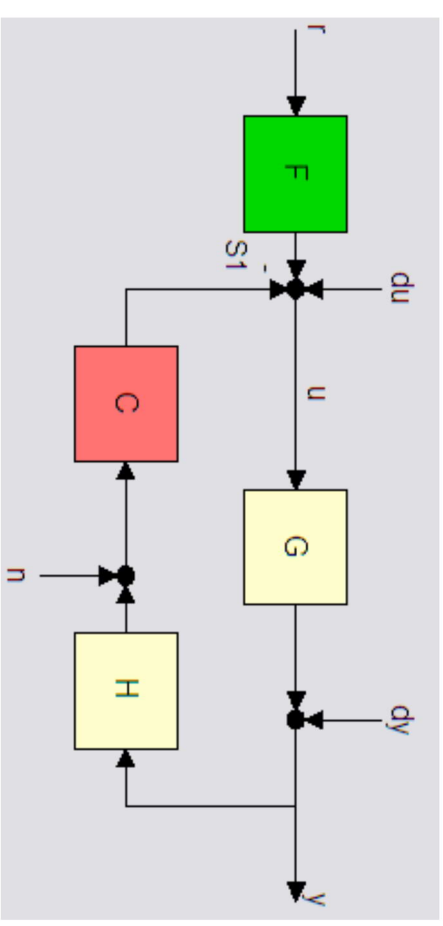
• First: $C_{y \rightarrow u}$

- reaction to disturbances
- sensitivity to change G_{ob}
- compromise between speed and robustness
- $C_{r \rightarrow u}$ less critical



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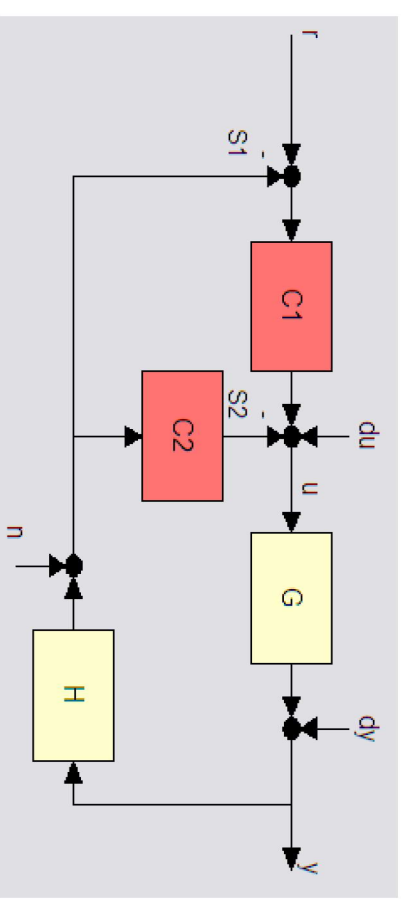
Several structures of the 2DOF controller



- basic theoretical structure,
- not implementable for an integral controller
- would require the subtraction of two large numbers, which will result in numerical errors

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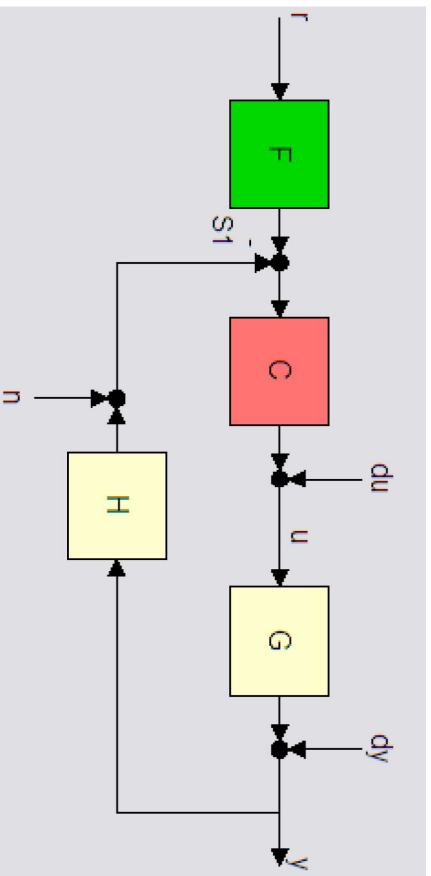
Several structures of the 2DOF controller



- a very popular structure in industrial controllers
- the C1 part contains the PI action - eliminates the steady state deviation
- the C2 part contains the PD action - fast response to disturbances and load

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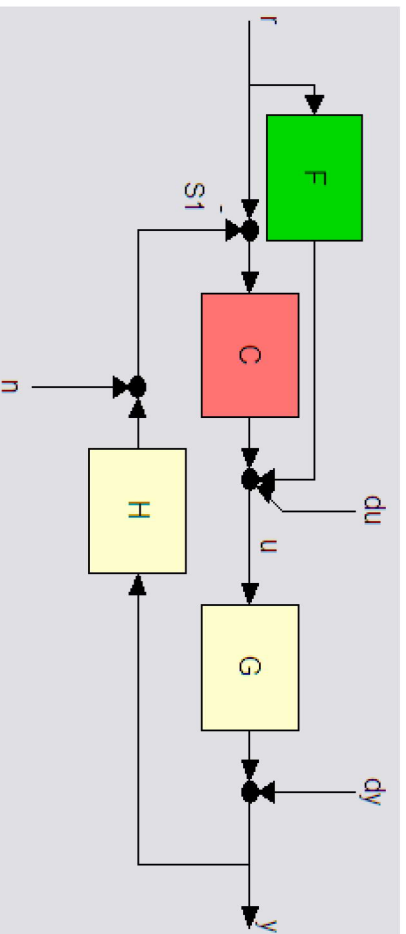
Several structures of the 2DOF controller



- structure with pre-processing of the reference signal
- may include, for example, filters that do not pass frequencies that cause mechanical vibration - input shaping
- typical applications - precision drives

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Several structures of the 2DOF controller



- structure with feed forward of the reference signal
- may include, for example, may include, for example, an inverse dynamics model of an industrial robot
- typical applications - fast, high-dynamic drives

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ISA: International Society of Automation

Standard forms of PID controller - applications in industrial controllers..

$$E(s) = Y_{sp}(s) - Y(s)$$

Formula I:

$$U = K \left(b Y_{sp} - Y + \frac{1}{s T_i} E + \frac{s T_d}{1 + s T_d / N} (c Y_{sp} - Y) \right)$$

Weighting coefficients of the reference signal

Different definitions of error for parts P, I, D.

$$e_p = b y_{sp} - y$$

$$e_d = c y_{sp} - y$$

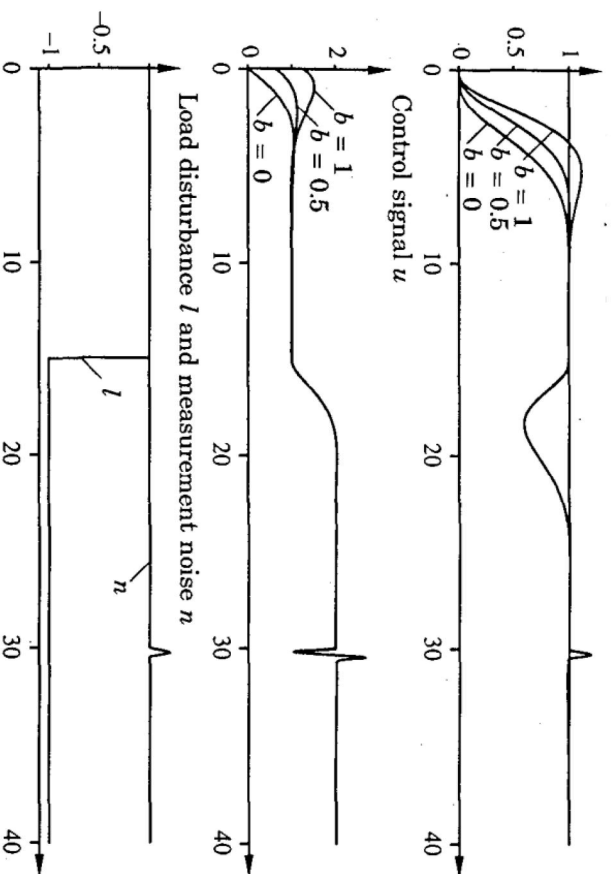
In practice, the most common are

$$e = y_{sp} - y$$

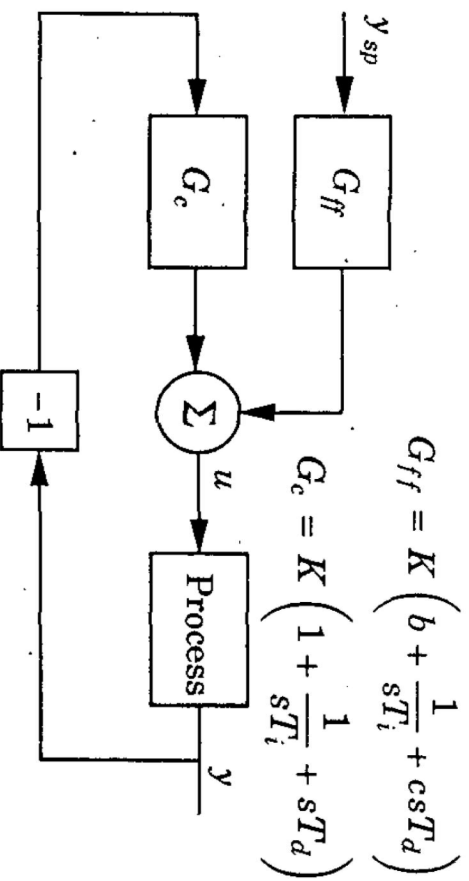
$c = 0$

Coefficients b and c influence the system response to the set signal without changing the disturbance response

The effect of the weight coefficient b



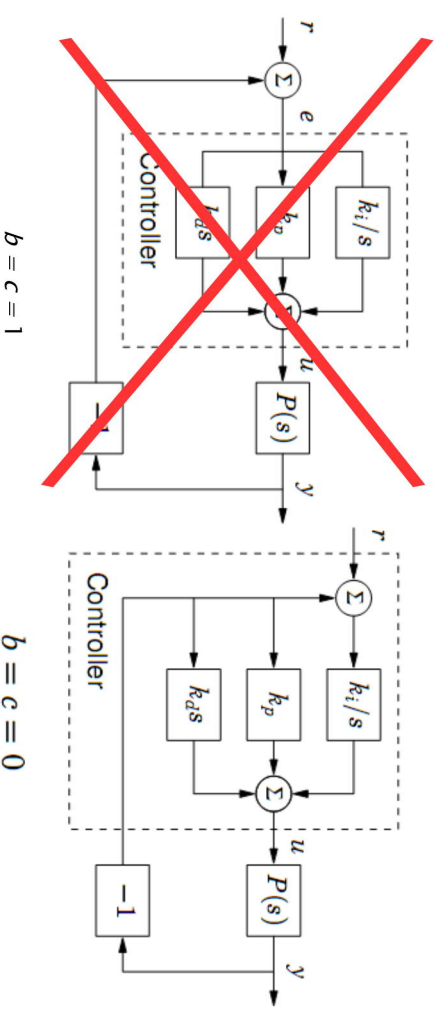
2 DOF controller



$b=1$ and $c=0$ – PI-D
 $b=0$ and $c=0$ – I-PD

(‘Poor man’s 2DOF’*)

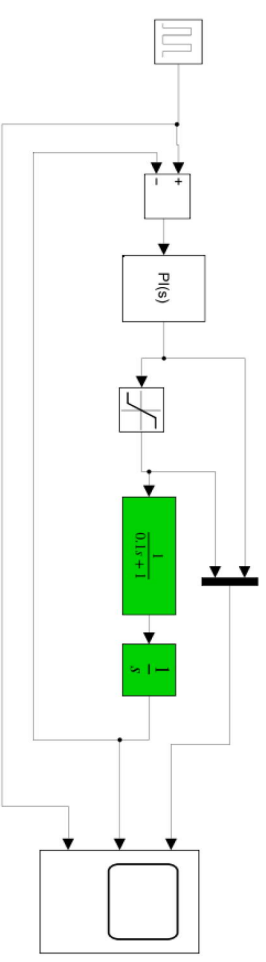
$$U(s) = k_p (bR(s) - Y(s)) + \frac{k_i}{s} (R(s) - Y(s)) + k_d s (cR(s) - Y(s))$$



*) B. Bernhardsson, K.J. Åström Control System Design – PID Control, 2007

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Wind-up problem

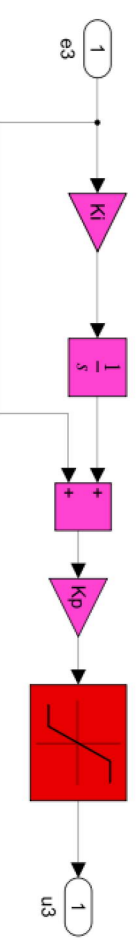


Ideal form of controller

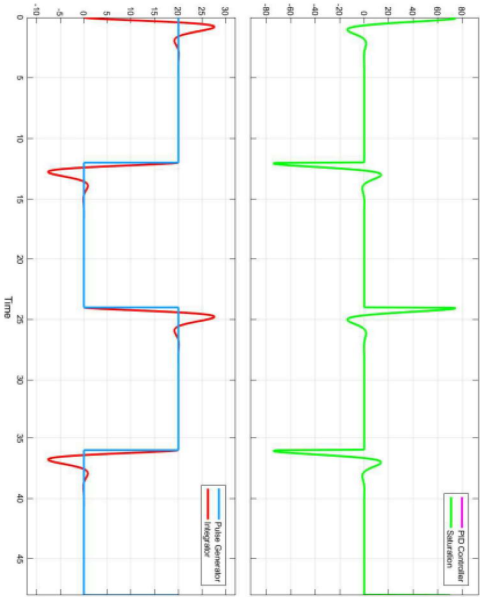
$$P = 3.5$$

$$I = 2$$

$$G_{PI}(s) = P \left(1 + I \frac{1}{s} \right)$$



Wind-up problem - system without saturation

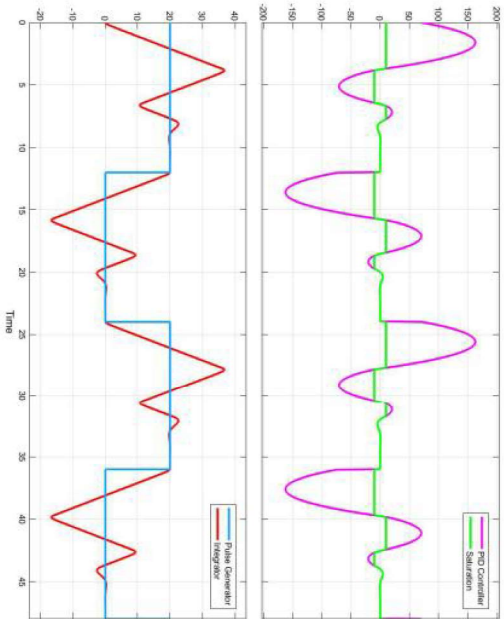


In saturation block:

upper limit = inf

lower limit = -inf

Wind-up problem - system with saturation

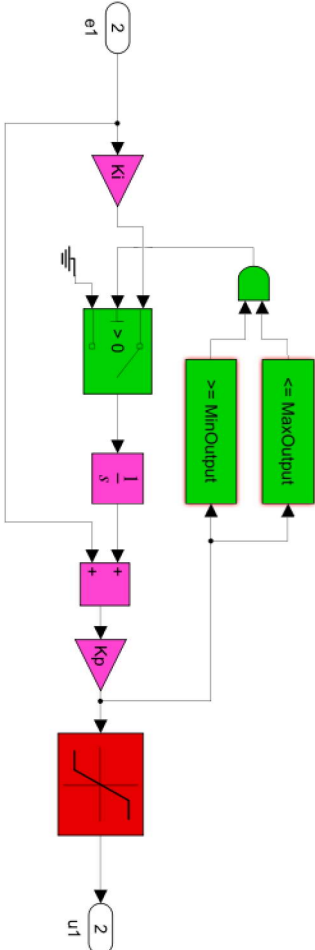


In saturation block:

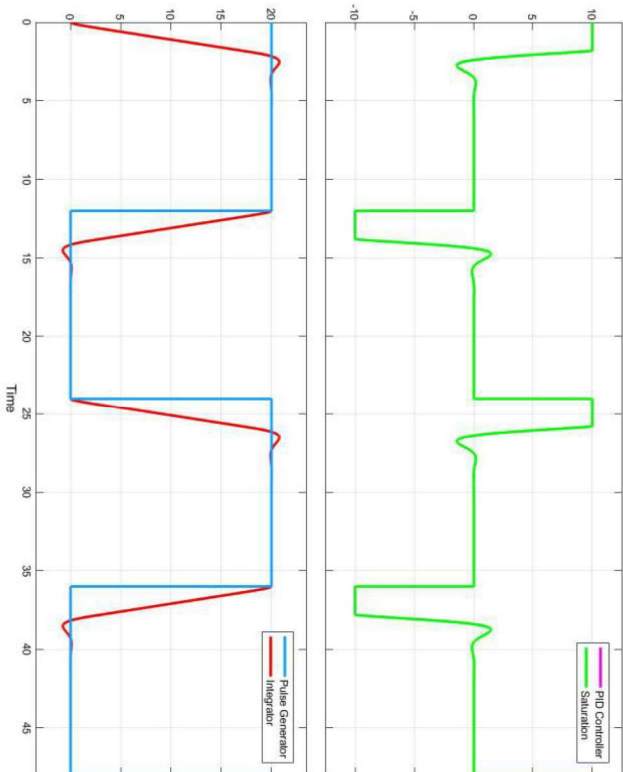
upper limit = 10

lower limit = -10

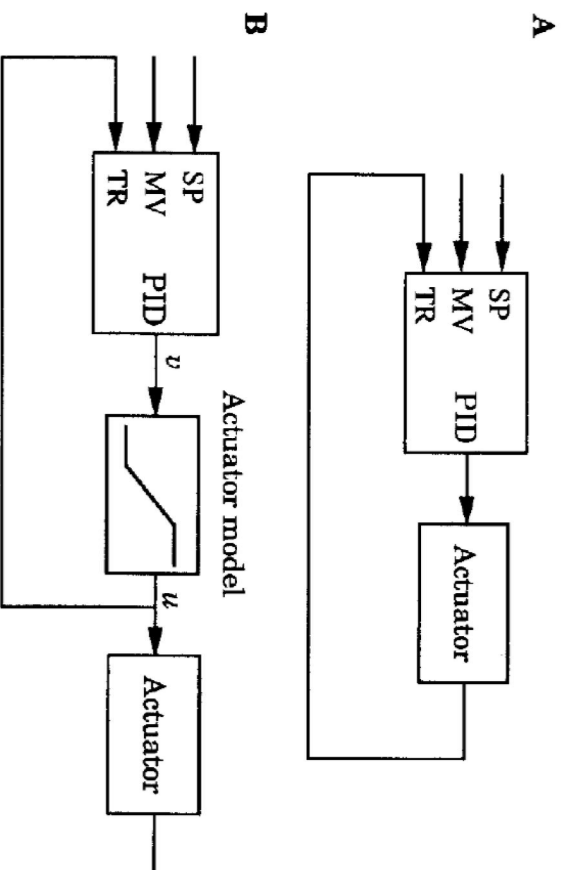
Anti Wind-up (AWU) solution - clamping



Anti Wind-up (AWU) solution - clamping



Industrial PID controller



S7-1200

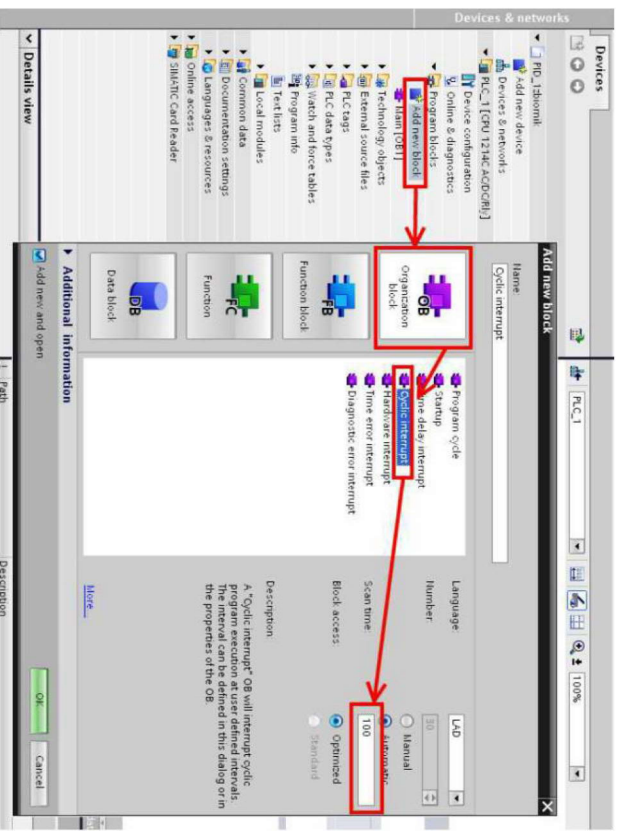


Types of PID blocks in the S7-1200

- **PID_Compact** - universal PID controller designed for process control
- **PID_3Step** - for process control with actuators controlled by on and off signals (e.g. motorised valves)

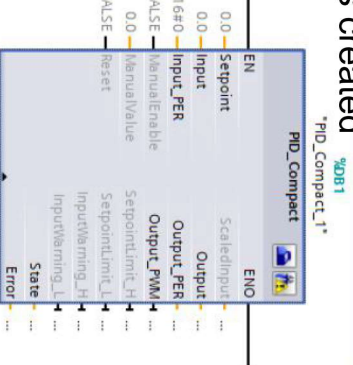
Tuning of the controller

- User selected
- Selected by a self-tuning process (autotuning):
 - "pretuning" mode: based on rise time and dead time,
 - "fine tuning" mode: based on fixed and limited oscillations (relay method).



The controller function block must be called with a fixed sampling period - for example in a cyclic interrupt

A new instance (with its own data) of the controller function block is created



Name	Description	Version
PID Control		V1.0
CompactPID		V1.2
PID_Compact	Universal PID controller...	V1.2
PID_3Step	PID controller with tuni...	V1.1
Motion Control		

PID Compact

$$y = K_p \left[(b \cdot w - x) + \frac{1}{T_i \cdot s} (w - x) + \frac{T_d \cdot s}{a \cdot T_d \cdot s + 1} (c \cdot w - x) \right]$$

